Understanding Understanding

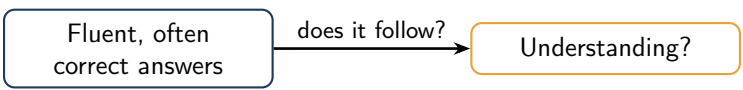
A Structural Account Across Human and Machine Cognition

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Motivation: a question with new urgency

- Understanding is invoked constantly in everyday and philosophical discourse, yet rarely defined with precision
- Large-scale machine learning systems now answer difficult questions fluently and often correctly
- Do these systems understand what they are doing?
- Or have they found a way to perform well without any of the properties that make human understanding valuable?



Central claim

Thesis

Understanding is best treated not as a distinctive mental state or as a species of propositional knowledge, but as a **structural achievement**.

- The reliable ability to navigate a space of **implications**, **counterfactuals**, and **transformations** surrounding a domain
- Not merely the ability to produce one correct output for one input

Not a distinctive mental state

Not a species of propositional knowledge

A structural achievement

Strand 1: understanding differs from knowledge

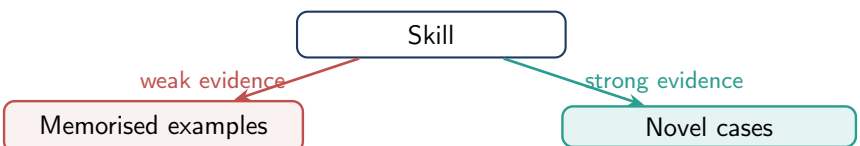
- The philosophical case for distinguishing understanding from knowledge (Kvanvig, 2003; Elgin, 2017; Grimm, 2006)
- Understanding tolerates idealisation
- Understanding resists reduction to any finite list of true propositions

Implications

An account of understanding cannot simply count how many correct propositions an agent holds or can output.

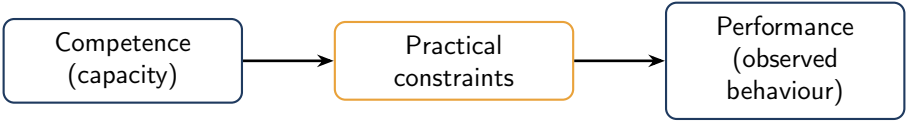
Strand 2: the ability view

- Understanding as a cluster of cognitive skills (Ryle, 1949; Hills, 2016; de Regt, 2017)
- Tested by generalisation to novel cases
- Not tested by reproduction of memorised examples

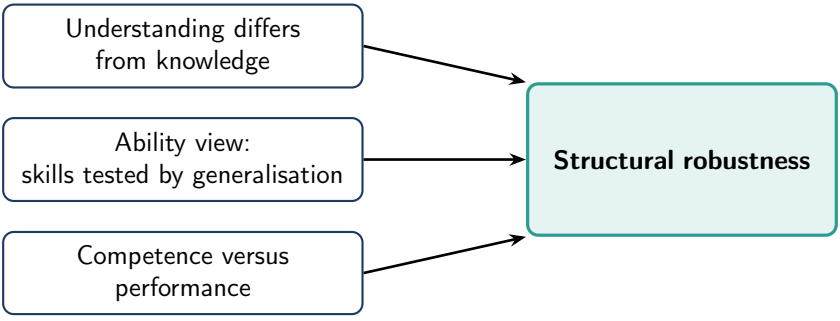


Strand 3: competence and performance

- A distinction from cognitive science (Chomsky, 1965; Firestone, 2020)
- Separates an agent's underlying capacity from its observed behaviour under practical constraints
- A single observed output may reflect the capacity, or the constraints under which it was exercised

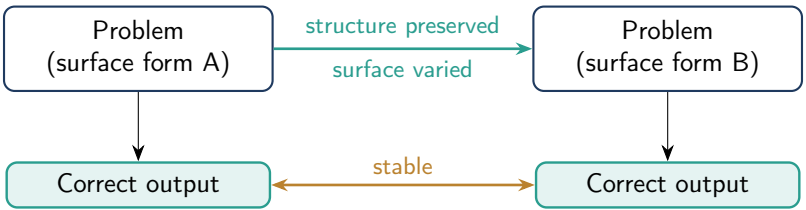


Three strands, one proposal



Definition (informal)

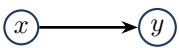
An agent understands a domain to the extent that its correct performance is **stable under transformations that preserve the domain's underlying structure** while varying its surface features.



Generalises the abstraction consistency criterion used in benchmarks such as the Abstraction and Reasoning Corpus (Chollet, 2019).

Performance versus understanding

Performance only
one input, one correct output



beyond

Space surrounding the domain

Implications

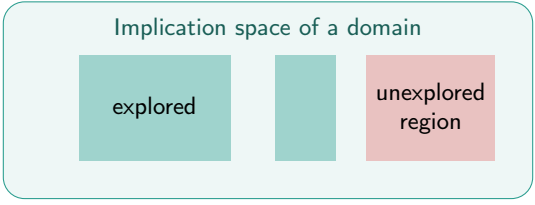
Domain

Counterfactuals

Transformations

Failures of understanding: structured ignorance

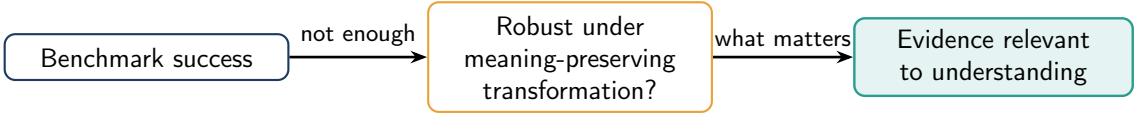
- Failures of understanding are frequently failures of **structured ignorance**
- That is, failures to explore particular regions of a domain's implication space
- Not simple absences of information
- Clear analogues in heuristics and biases research (Tversky and Kahneman, 1974)



Applying the framework to machines

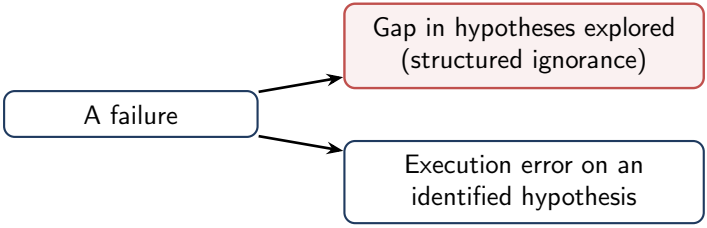
Claim

Superhuman performance on a reasoning benchmark is not, by itself, evidence of understanding.



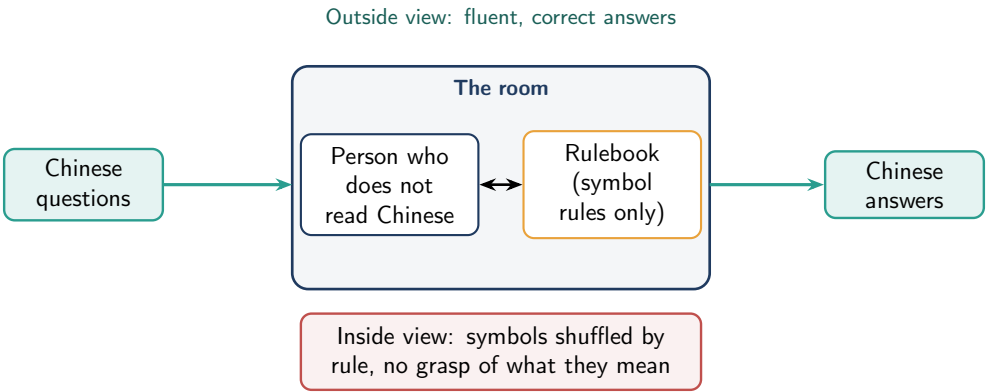
Two diagnostic questions for machine understanding

1. Is performance robust under meaning-preserving transformation?
2. Do failures reflect genuine gaps in the hypotheses explored, or execution errors on hypotheses already identified?



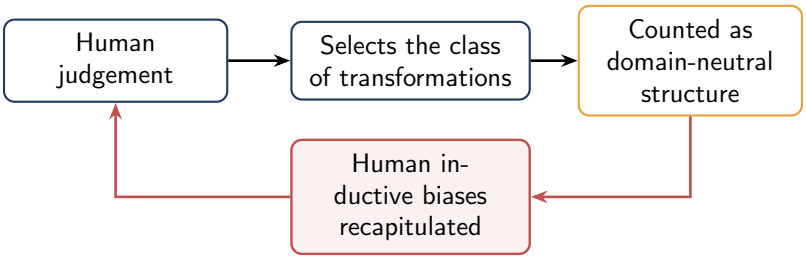
Objection: the Chinese Room

- The worry: behavioural robustness cannot settle questions of semantic content (Searle, 1980)
- Robust behaviour alone leaves the question of semantic content unanswered



A further concern: anthropocentrism

- The class of transformations counted as meaning-preserving is itself selected by human judgement
- Risk: recapitulating human inductive biases under the guise of domain-neutral structure
- Draws on the argument of Holm and Banerjee (2024) that comparisons of intelligence across humans, animals, and machines are already anthropocentric



What remains open

- **Metaphysical:** is structural robustness, however extensive, sufficient for the fullest sense of understanding, or only a necessary condition and reliable signature of it?
- **Specification:** how can the relevant transformations be specified without introducing a human-centred standard?
- **Empirical:** how should structural robustness be measured in both human and artificial reasoners?

Deliberate openness

Progress on the empirical question can proceed productively even while the other questions remain unsettled.

Conclusions

- Understanding is a structural achievement: reliable navigation of a domain's implications, counterfactuals, and transformations
- Structural robustness operationalises this: stable correct performance under structure-preserving transformations
- Failures are often structured ignorance, not mere missing information
- Superhuman benchmark performance is not, by itself, evidence of machine understanding
- The class of meaning-preserving transformations is human-selected, and specifying it without a human-centred standard is an open problem

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